with endolymph. At the base of the cochlea, the scala vestibule ends at the 'oval window' whereas the scala tympani ends at the 'round window' of the middle ear. The chambers scala vestibuli and scala media are separated by a membrane called Reisner's membrane whereas the scala media and scala tympani are separated by a membrane called Basilar membrane (Figure 10.17)

Organ of Corti

The **organ of Corti** (Figure.10.18) is a sensory ridge located on the top of **the Basilar membrane** and it contains numerous hair cells that are arranged in four rows along the length of the basilar membrane. Protruding from the apical part of each hair cell is hair like structures known as **stereocilia.** During the conduction of sound wave, stereocilia makes a contact with the stiff gel membrane called **tectorial membrane,** a roof like structure overhanging the organ of corti throughout its length.

Mechanism of hearing

Sound waves entering the external auditory meatus fall on the tympanic membrane. This causes the ear drum to vibrate, and these vibrations are transmitted to the oval window through the three auditory ossicles. Since the tympanic membrane is 17-20 times larger than the oval window, the pressure exerted on the oval window is about 20 times more than that on the tympanic membrane. This increased pressure generates pressure waves in the fluid of perilymph. This pressure causes the round window to alternately bulge outward and inward meanwhile the basilar membrane along with the organ of Corti move up and down. These movements of the hair alternately open and close the mechanically gated ion channels in the base of hair cells and the action potential is propagated to the brain as sound sensation through cochlear nerve.

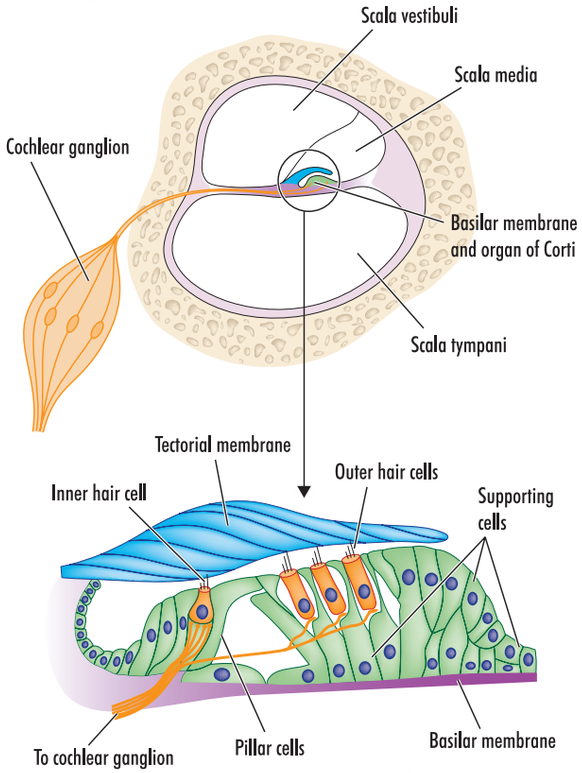


Figure 10.18 Organ of Corti

Defects of Ear

Deafness may be temporary or permanent. It can be further classified into **conductive deafness** and **sensory-neural deafness**. Possible causes for conductive deafness may be due to

i. the blockage of ear canal with earwax,
ii. Rupture of eardrum
iii. Middle ear infection with fluid accumulation
iv. Restriction of ossicular movement.In **sensory-neural deafness**, the defect may be in the organ of Corti or the auditory nerve or in the ascending auditory pathways or auditory cortex.

Organ of Equilibrium

Balance is part of a sense called proprioception, which is the ability to sense

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